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Paleomagnetism and U-Pb Geochronology of Early Cretaceous Volcanic Rocks from the Qiangtang Block, Tibetan Plateau: Implications for the Qiangtang-Lhasa Collision

Yong Cao[a], Zhiming Sun[a,*], Haibing Li[b], Xiaozhou Ye[a], Jiawei Pan[b], Dongliang Liu[b], Lei Zhang[b], Bailing Wu[a], Xinwen Cao[a], Chenguang Liu[a], Zhenyu Yang[c]

[a] Key Laboratory of Paleomagnetism and Tectonic Reconstruction of Ministry of Natural Resource, Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China
[b] Key Laboratory of Deep-Earth Dynamics of Ministry of Natural Resources, Institute of Geology, Chinese Academy of Geological Sciences, Beijing 100037, China
[c] College of Resources, Environment and Tourism, Capital Normal University, Beijing 100048, China

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ABSTRACT

The collision between the Qiangtang and Lhasa blocks is a critical factor in understanding the geodynamics of the central Tibetan Plateau. However, the process of the Qiangtang-Lhasa collision remains contentious. A direct way to study the history of the Qiangtang-Lhasa collision would be to determine the evolution of the paleolatitude positions of the Qiangtang and Lhasa blocks during the Jurassic-Cretaceous. In this study, we present a combined paleomagnetic and geochronological study of the Early Cretaceous volcanic rocks dated at ~120–115 Ma in the western Qiangtang block. Stepwise thermal demagnetization succeeded in isolating the high-temperature characteristic directions of the samples. The tilt-corrected mean direction of the 16 sampling sites was $Ds = 60.9^{\circ}$, $Is = 45.9^{\circ}$ with $a_{95} = 4.4^{\circ}$, which indicates that the Qiangtang block was situated at $27.6 \pm 5.0^{\circ}$N during the Early Cretaceous (~120–115 Ma) (reference point: 32.9°N, 83.5°E). Our new data combined with previous reliable Cretaceous paleomagnetic results for the Lhasa block indicate that the collision between the western parts of the Qiangtang and Lhasa blocks may occur later than 115 Ma. The extent of the western segment of the Bangong-Nujiang Tethys Ocean was $\sim 825 \pm 600$ km ($7.5^{\circ} \pm 5.5^{\circ}$ in latitude) at 120–115 Ma.

1. Introduction

The Tibetan Plateau is comprised of a series of blocks (Fig. 1) and was formed by the collision-amalgamation of these blocks (Yin and Harrison, 2000). The collisional process and subsequent effects played a crucial role in studies of the formation and evolution of the Tibetan Plateau. The Qiangtang and Lhasa blocks are located in the central Tibetan Plateau (Fig. 1). The two blocks drifted away from Gondwana and collided with the southernmost part of Asia in the Mesozoic (Zhu et al., 2016). The collision between the Qiangtang and Lhasa blocks was accompanied by the closure of the Bangong-Nujiang Tethys Ocean (BNTO), which created the Bangong-Nujiang Suture Zone (BNSZ) and may have resulted in a large amount of crustal shortening. This exerted a significant influence on the evolution of the central Tibetan Plateau prior to the India-Asian collision (Kapp et al., 2007; Murphy et al., 1997). Therefore, the collisional processes between the Qiangtang and Lhasa blocks is a key factor in understanding the evolution and geodynamics of the central Tibetan Plateau.

Despite the fact that an increasing number of studies have focused on the collision between the Qiangtang and Lhasa blocks, the timing of the Qiangtang-Lhasa collision remains contentious. The proposed age of the Qiangtang-Lhasa collision (the closure of the BNTO) ranges from the Middle Jurassic to the Late Cretaceous and is mainly based on stratigraphic and sedimentological records (Kapp et al., 2007; Sun et al., 2019), ophiolitic fragments along the BNSZ (Zhang et al., 2012; Liu et al., 2014), collision-related magmatism (Li et al., 2015), paleomagnetism (Bian et al., 2017; Chen et al., 2017), paleontology (Baxter et al., 2009), and synthetic analysis (Yin and Harrison, 2000; Zhu et al., 2016).

Similar to the India-Asia collision, the assembly of the Qiangtang and Lhasa blocks was the result of a nearly north-south continent-continent collision. Thus, it is crucial to determine the Cretaceous

* Corresponding author.
E-mail address: sunzm1209@163.com (Z. Sun).

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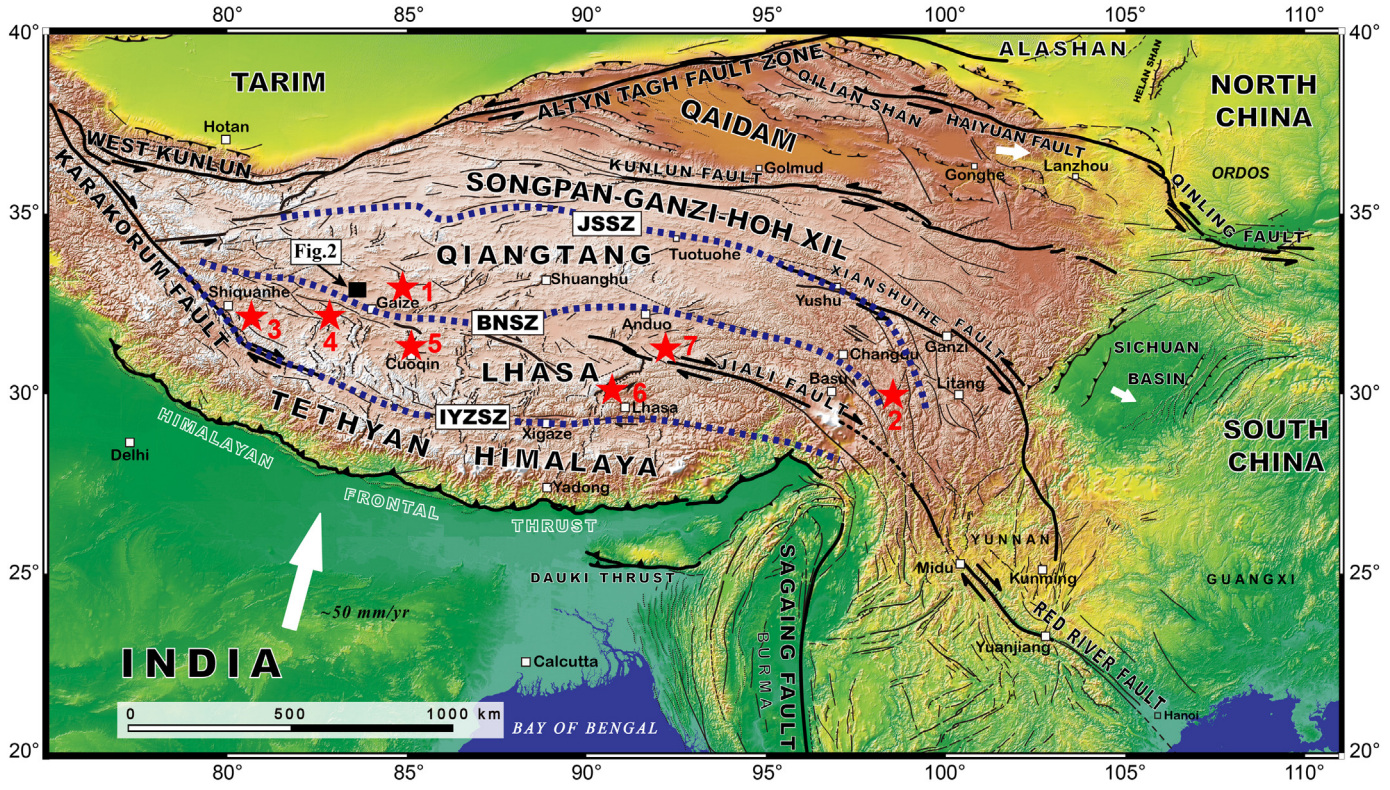


Fig. 1. Tectonic map of the Tibetan Plateau and its adjacent regions (modified after Tapponnier et al., 2001). IYZSZ: Indus-Yarlung Zangbo Suture Zone; BNSZ: Bangong-Nujiang Suture Zone; JSSZ: Jinsha Suture Zone. Red stars: distribution of previous paleomagnetic sampling locations in the Early Cretaceous strata of the Qiangtang and Lhasa blocks, 1-Chen et al. (2017), 2-Otofuji et al. (1990) and Huang et al. (1992), 3-Bian et al. (2017), 4-Ma et al. (2014), 5-Chen et al. (2012), 6-Sun et al. (2008), and 7-Li et al. (2017). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

paleolatitudinal positions of the two blocks in order to understand the collision between the Qiangtang and Lhasa blocks. Previous Cretaceous paleomagnetic studies of the Lhasa block have provided important constraints on its paleolatitudinal positions (Chen et al., 2012; Sun et al., 2012; Yang et al., 2015; Bian et al., 2017; Cao et al., 2017; Li et al., 2017). However, little reliable Early Cretaceous paleomagnetic data are available for the Qiangtang block (Otofuji et al., 1990; Huang et al., 1992; Chen et al., 1993; Chen et al., 2017). Furthermore, many of these paleomagnetic results are defined by a small number of samples, have provided no positive field test, and/or lack accurate geochronologic constraints. In addition, the southeastern Qiangtang block is located close to the eastern Himalayan syntaxis, and it may have been extruded southeastward as a result of the collision between the Indian and Asian continents (Tapponnier et al., 2001; Replumaz and Tapponnier, 2003; Otofuji et al., 2010). This consecutive extrusion might have been accompanied by a large-scale tectonic rotation of the eastern Qiangtang block (Tapponnier et al., 2001; Otofuji et al., 2010; Tong et al., 2015). Therefore, the paleomagnetic results for the eastern part of the Qiangtang block (Otofuji et al., 1990; Huang et al., 1992) may not be suitable for tectonic analysis of the western part of the Qiangtang block.

The foregoing review demonstrates that there are significant limitations in discussing the collisional process between the Qiangtang and Lhasa blocks using earlier paleomagnetic studies of the Qiangtang block. In this study, we present a combined paleomagnetic and U—Pb geochronological study of the Early Cretaceous volcanic rocks from the western part of the Qiangtang block to provide reliable paleomagnetic constraints on the collisional process between the Qiangtang and Lhasa blocks.

2. Geologic setting and sampling

The Qiangtang block is one of the main blocks of the Tibetan Plateau (Fig. 1). The width of the Qiangtang block is 500–600 km in the central part and < 150 km in both the eastern and western parts. The Qiangtang block is separated from the Lhasa block by the BNSZ to the south and from the Songpan-Ganzi-Hoh Xil block by the Jinsha Suture Zone (JSSZ) to the north (Fig. 1) (Yin and Harrison, 2000). Our study area is located about 90 km to the northwest of Gaize County. The outcrops in this area are mainly Jurassic and Cretaceous strata, including the Quse Formation (J_1q), the Sewa Formation (J_2s), the Meiriqieco Formation (K_1m), and the Abushan Formation (K_2a) (Fig. 2a). The Sewa Formation mainly consists of gray siltstone. The Abushan Formation mainly consists of conglomerates and conglomeratic sandstones. The Sewa Formation is unconformably overlain by the Early Cretaceous Meiriqieco Formation, implying that the folding of the Sewa Formation occurred earlier than the Early Cretaceous.

Paleomagnetic samples were collected from the volcanic rocks of the Lower Cretaceous Meiriqieco Formation at 32.9°N, 83.5°E, about 50 km north of the BNSZ. The Meiriqieco Formation, which is mainly composed of purple-red andesite interbedded with gray tuff (Fig. 2b), unconformably overlies the Middle Jurassic (Sewa Formation) and is unconformably overlain by the Upper Cretaceous (Abushan Formation). The attitude of the stratum is clearly illustrated by the petrographic boundary between the andesite and tuff (Fig. 2c, d) and can be used for structural correction in the paleomagnetic study. The Upper Cretaceous Abushan Formation has an approximately horizontal dip and unconformably overlies the Meiriqiecuo Formation, indicating that the tilting of the Meiriqiecuo Formation occurred from the late of the Early Cretaceous to the Late Cretaceous.

A total of 164 samples were collected from 17 paleomagnetic sampling sites for paleomagnetic investigations of the volcanic rocks of

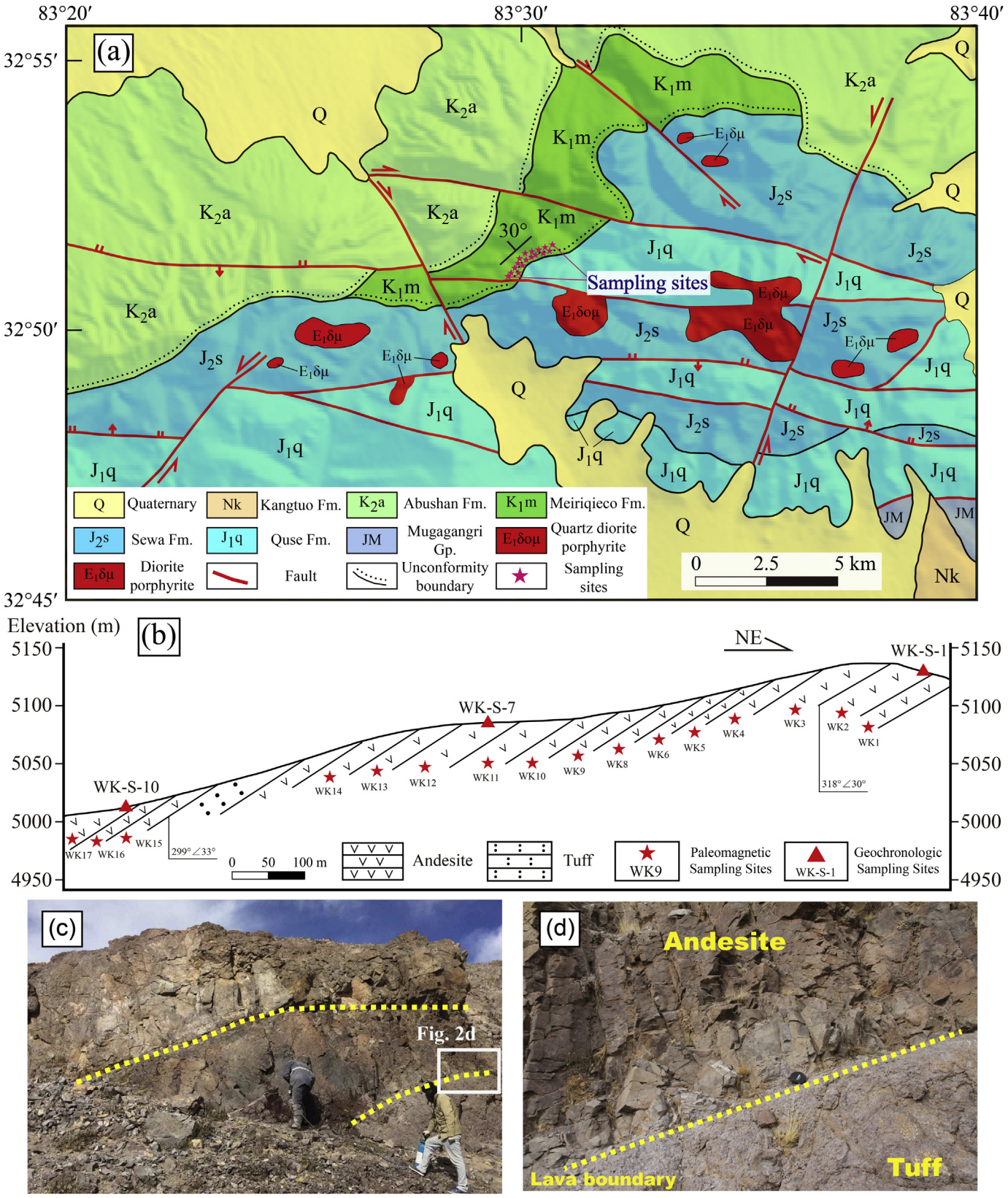


Fig. 2. (a) Geologic map of the study area (modified from the Wuma regional geological investigation report). (b) Simplified geological section of the sampling area. (c) and (d) Photographs showing field outcrops and the lava boundary of the volcanic rocks of the Meiriqieco Formation.

the Early Cretaceous Meiriqieco Formation using a gasoline-powered drill. The studied section has distinct volcanic rocks, which have significantly different colors and structures. Each sampling site included at least one discrete cooling unit. The core samples were oriented using both magnetic and sun compasses. In addition, three fresh samples of andesite from the upper (WK-S-10), middle (WK-S-7), and lower (WK-S-1) parts of the study section were collected for isotopic geochronologic analysis. All of the drill cores were cut into standard specimen dimensions (2.2-cm-long cylinders) in the laboratory for paleomagnetic analysis.

3. Experiments

The U−Pb dating of the zircon grains was conducted using a Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS) at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences, Wuhan, China. Detailed measurement procedures for the laser ablation system and the ICP-MS instrument are described by Liu et al. (2010).

Representative specimens underwent thermal demagnetization of three-component Isothermal Remanent Magnetization (IRM) (Lowrie, 1990) and saturation Isothermal Remanent Magnetization (SIRM) acquisition and demagnetization curves using a JR-6A spinner magnetometer. Thermal demagnetization of the three-component IRM used fields of 0.12 T, 0.4 T, and 2.5 T. The thermal demagnetization of each specimen involved 15 to 20 steps up to a maximum temperature of 680 °C, using an ASC TD-48 oven, combined with a JR-6A spinning magnetometer. The measurements were performed in a shielded room at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Natural Resource, China. The hysteresis loops and a first-order reversal curves (FORC) were obtained using a Princeton Vibrating Sample Magnetometer (MicroMag™ Model 3900 VSM). The magnetization directions were determined using Principal Component Analysis (PCA) (Kirschvink, 1980), and the average paleomagnetic directions were calculated using Fisher statistics (Fisher, 1953).

Scanning electron microscope (SEM) observations were conducted on representative samples using the LEO Stereoscan 440 SEM at the Institute of Geology, Chinese Academy of Geological Sciences.

4. Results

4.1. U−Pb zircon dating

A total of 86 zircon grains were analyzed and their morphologies were mostly euhedral to subhedral with prismatic forms. The zircons exhibited typical oscillatory zoning indicating a magmatic origin (Fig. 3). The analyzed samples thorium (Th) contents ranged from 55.25 ppm to 919.64 ppm, and their U contents ranged from 104.4 ppm to 1501.29 ppm. Their Th/U ratios ranged from 0.38 to 1.22, with a mean Th/U ratio of 0.61. The U−Pb concordia ages of samples WK-S-1, WK-S-7, and WK-S-10 are 116.5 ± 1.7 Ma (mean square weighted deviation (MSWD) = 0.94), 115.9 ± 1.3 Ma (MSWD = 0.69), and 117.4 ± 1.3 Ma (MSWD = 1.00), respectively (Fig. 4). Previous U−Pb zircon ages for the volcanic rocks of the Early Cretaceous Meiriqieco Formation in the adjacent regions range from ~123 Ma to ~110 Ma (Ma et al., 2015; Wei et al., 2017). The U−Pb zircon dating results indicate that the Meiriqieco Formation andesite erupted during the Early Cretaceous.

4.2. Rock magnetic analyses

The IRM acquisition and demagnetization curves and the thermal demagnetization of a three-component IRM are illustrated in Fig. 5. The SIRM acquisition curves show that the remanence of the representative samples sharply increases below 200 mT, and slowly increases between 200 mT and 2.5 T. The remanences of the representative samples approached saturation by 2.5 T. The back-field demagnetization curves of the SIRM illustrate that the remanence of the representative samples rapidly decreases to zero at 40 to 50 mT (Fig. 5a, b). These findings indicate the presence of high- and low-coercivity magnetic minerals. The hard component of the representative samples showed an unblocking temperature of 680 °C, which is indicative of hematite. The thermal demagnetization of the medium and soft fractions revealed an unblocking temperature of 580 °C, which is indicative of magnetite (Fig. 5c, d).

The hysteresis loops, FORC diagrams, and Day plots of the hysteresis parameters of the representative samples are shown in Fig. 6. The hysteresis loops show that the samples were saturated at no more than 300 mT, indicating that low coercivity magnetic minerals are dominant. The low remanent coercive force (Hcr) and coercive field (Hc) values also indicate that the main magnetic minerals have a low coercivity. The wasp-waisted shapes of the hysteresis loops imply the presence of low-coercivity and high-coercivity magnetic minerals (Fig. 6a-b). The hysteresis parameters of the representative samples plot close to the multi-domain (MD) + single domain (SD) mixing curves (Fig. 6e) (Dunlop, 2002). The FORC diagrams with Hc values of ~15 mT are accordant with the distribution of the state particles (Fig. 6c-d). The state particles indicate a transitional state between the MD and SD (Roberts et al., 2017). The FORC result is consistent with the magnetic domain estimated from the Day plot. Both the FORCs and the Day plots indicate that the magnetic minerals fall within the SD + MD domain.

4.3. Paleomagnetic results

Representative orthogonal vector diagrams of the Early Cretaceous volcanic rocks of the Meiriqiecuo Formation are shown in Fig. 7. The thermal demagnetization results indicate that a low-temperature component (LTC) can be isolated before 180–470 °C for most of the samples (Fig. 7a, c-k). The site-mean LTC direction of the 10 sampling sites is $Dg = 7.2°$, $Ig = 41.3°$ with $kg = 16.6$, $a_{95} = 12.2°$ in geographic coordinates and $Ds = 353.1°$, $Is = 20.3°$ with $ks = 14.7$, $a_{95} = 13.0°$ in stratigraphic coordinates (Table S1, Fig. 8a). The values of κ and a_{95} decrease and increase after the tilt correction, respectively. The site-mean direction of the LTC in geographic coordinates is close to the present geomagnetic field (PGF) direction ($D/I = 1.4°/50.1°$) at the sampling locality (32.9°N, 83.5°E). Thus, the LTC of the volcanic rocks of the Meiriqiecuo Formation may be a viscous remanent magnetization of the PGF.

Site WK7 was discarded because the characteristic remanent magnetization (ChRM) could not be determined due to ambiguous or noisy orthogonal demagnetization diagrams. Except for site WK7, the stepwise thermal demagnetization succeeded in isolating the high-temperature component (HTC) of the other 16 sites. The HTC decays toward the origin and was determined between 300-470 °C and 680 °C for most of the samples. The site-mean HTC direction is $Dg = 82.7°$, $Ig = 31.0°$ with $kg = 58.3$, $a_{95} = 4.9°$ before tilt correction and $Ds = 60.9°$, $Is = 45.9°$ with $ks = 70.6$, $a_{95} = 4.4°$ after tilt correction (Table 1, Fig. 8b). The corresponding paleopole lies at 37.9°N, 162.2°E with $A_{95} = 5.1°$.

4.4. SEM observations

Combined with the rock magnetic results and the energy dispersive spectroscopy (EDS) analysis (Fig. 9f-k), this indicates that the iron oxide minerals are composed of magnetite and Ti-rich hematite. The Ti-rich hematite may have been produced by low-temperature oxidation (LTO) processes long after cooling (Geissman and Van der Voo, 1980). However, the SEM observations show that the iron oxide minerals in the studied rocks are irregular in shape (Fig. 9a-e) and the grain sizes of the magnetic minerals range from several micrometers to more than 100 μm. The boundaries between the magnetic minerals and the groundmass (feldspar) or other minerals (e.g., apatite) are very distinct.

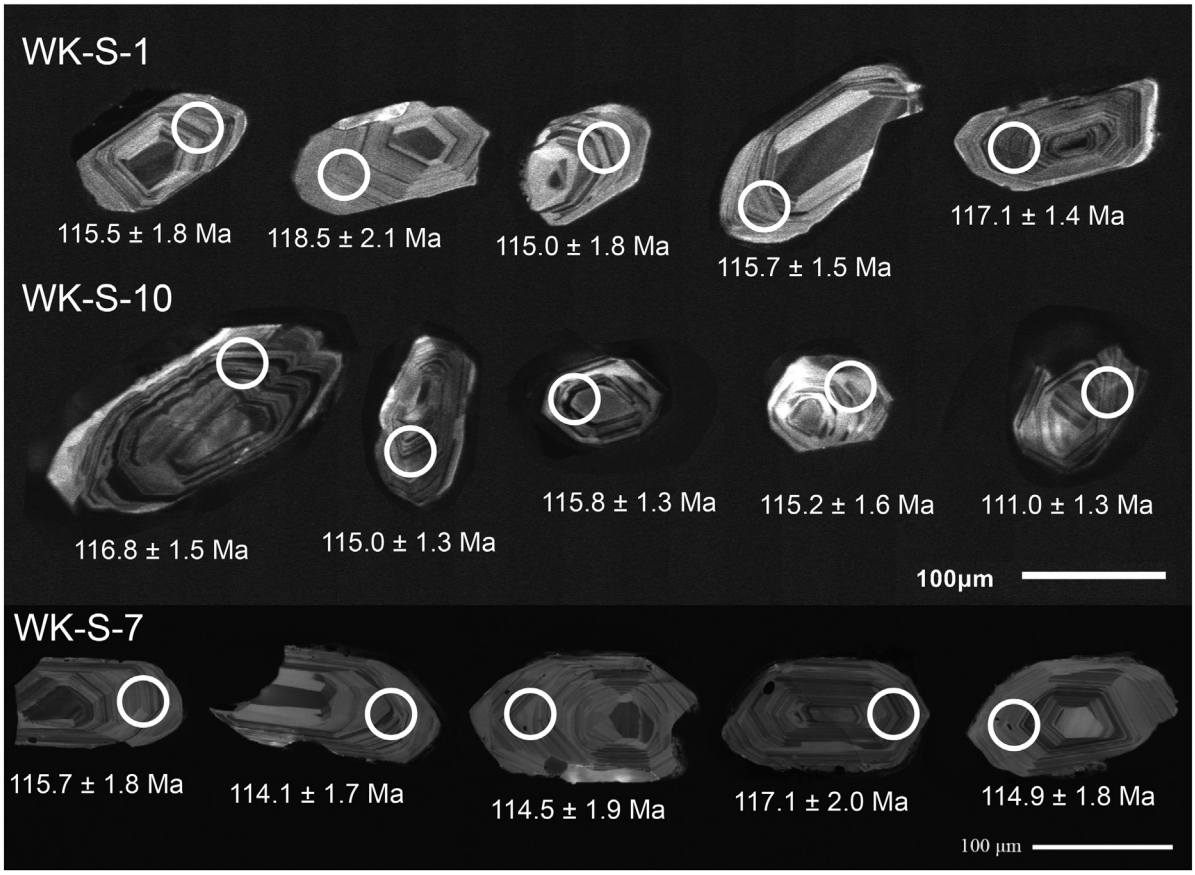


Fig. 3. Cathodoluminescence (CL) images and corresponding $^{206}Pb/^{238}U$ analysis results for representative zircons from the Early Cretaceous volcanic rocks in the Qiangtang block.

Oxidized rims were not present around the magnetic minerals (Fig. 9a, b). These textures indicate that the magnetic minerals should be primary and have not been affected by chemical alteration or weathering. Furthermore, the thermal demagnetization results indicate that the ChRM of the magnetite and hematite are almost similar. The above lines of evidence suggest that the iron oxide minerals were produced by exsolution during initial cooling, rather than by LTO.

5. Discussion

5.1. Reliability of the paleomagnetic directions

In order to test whether the paleomagnetic directions obtained in this study averaged out the paleosecular variation (PSV) of the geomagnetic field, an analysis of the angular dispersion of the virtual geomagnetic poles (VGPs) was performed for the polarity groups. The age of the studied section ranges from ~120 Ma to ~115 Ma. This age falls within the Cretaceous Normal Polarity Superchron (CNS: ~125–84 Ma) (Ogg, 2012; Biggin et al., 2008). The VGP scatter of the HTC directions obtained is 11.1°, which is much less than the predicted VGP scatter before the CNS for paleolatitudes of 0°-20° (S_λ = 18.5°, S_L = 16.6°, S_U = 21.0°) and for paleolatitudes of 30°-40° (S_λ = 18.5°, S_L = 16.5°, S_U = 21.1°) (McFadden et al., 1991). However, the VGP scatter is consistent with the expected value during the CNS for paleolatitudes of 10°-25° (S_λ = 8.7°, S_L = 7.0°, S_U = 11.3°) and for paleolatitudes of 35°-40° (S_λ = 12.2°, S_L = 10.1°, S_U = 15.4°) (McFadden et al., 1991). Moreover, the VGP scatter falls within the limit of the predicted value (9.9°-13.7°) at 27.3°N during the CNS (Biggin et al., 2008). However, the use of the VGP scatter to evaluate the time averaging of the PSV is controversial (Linder and Gilder, 2012). The VGP scatter may be a mathematical aberration resulting from the conversion of directions to poles (Linder and Gilder, 2012). Therefore, the angular standard deviation (ASD) of the observed VGPs was calculated by using the method of Vandamme (1994). The ASD of the observed VGPs is 11.9°, which is consistent with the predicted value for the global PSV models during the last 5 Ma at a similar latitude (Johnson et al., 2008). Furthermore, the upper A_{95max} and lower A_{95min} limits are 4.0° and 14.3°, respectively, according to the model of the "N-dependent A_{95} envelope" proposed by Deenen et al. (2011). The value of A_{95} is 5.1° for the VGPs of the HTC directions (Table 1), which falls within the reliability interval defined by Deenen et al. (2011). Therefore, the overall mean direction of the HTC of the Early Cretaceous volcanic rocks investigated in this study may have fully averaged out the PSV of the geomagnetic field.

The mean HTC direction is Dg = 82.7°, Ig = 31.0° with kg = 58.3, a_{95} = 4.9° before the tilt correction and is Ds = 60.9°, Ig = 45.9° with kg = 70.6, a_{95} = 4.4° after the tilt correction. The value of κ was increased ($\kappa g/\kappa s$ = 0.83) and the value of a_{95} was decreased ($a_{95}g/a_{95}s$ = 1.1) by the tilt correction. The monoclinal sequence of the Early Cretaceous volcanic rocks do not allow the fold test to be applied with any significance. However, the Watson and Enkin (1993) test gives a maximum grouping of directions at 117.5% unfolding (with uncertainties ranging from 92.5% to 139.9%). Furthermore, as listed in Table 1, only normal polarity is observed at all sampling sites, which is consistent with deposition and magnetization within the CNS with an age between 125 Ma and 84 Ma. Based on the analysis of the time averaging of secular variation, the overall mean direction of the HTC of the Early Cretaceous volcanic rocks may have fully averaged out the PSV of the geomagnetic field. In addition, the SEM observations indicate that the rocks in the studied sections contain a typical magmatic structure and have not been influenced by alteration. Based on the above analysis, the HTC magnetization most likely represents the

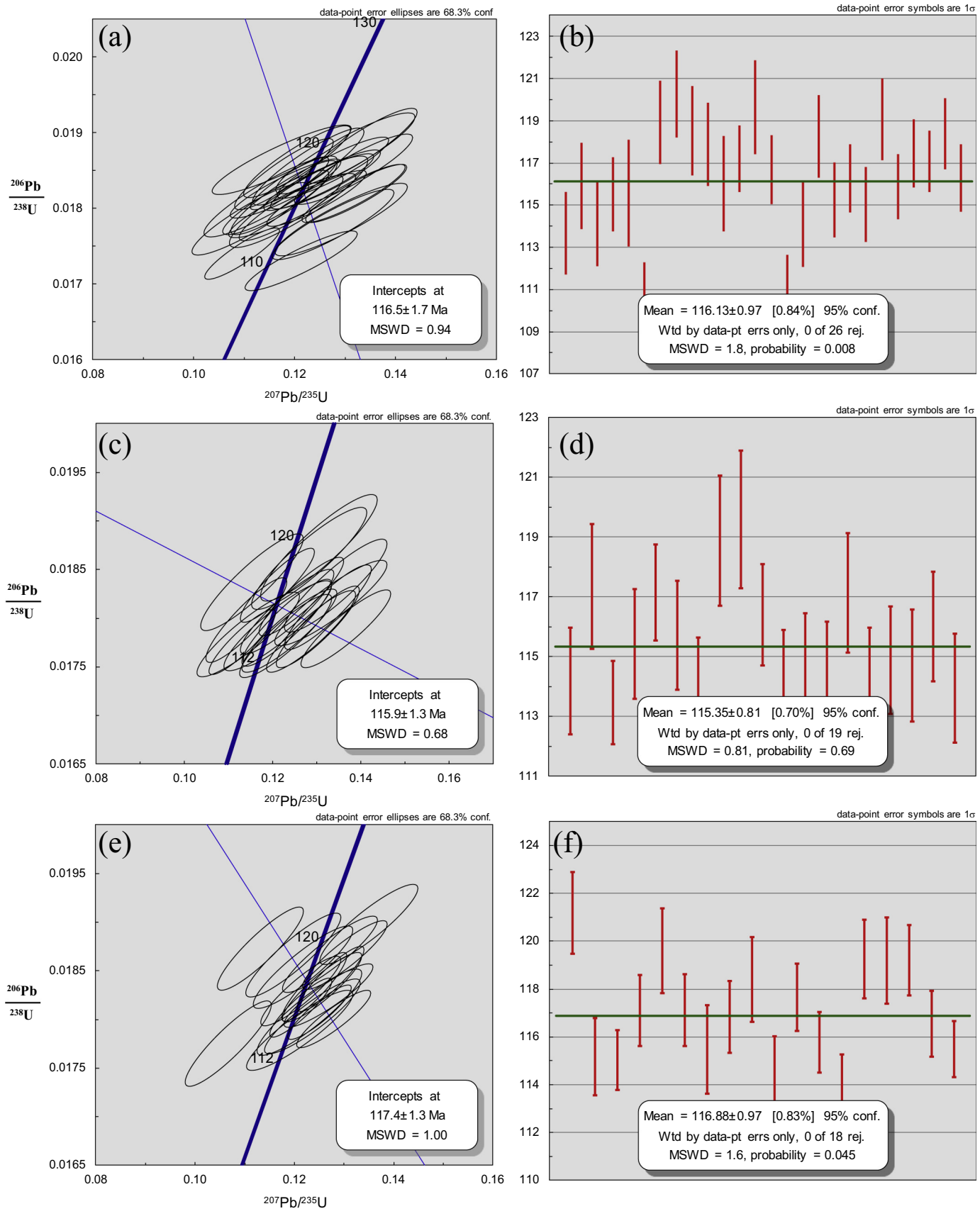


Fig. 4. (a, c, and e) U–Pb concordia diagram and (b, d, and f) weighted mean ages of zircons from the Early Cretaceous volcanic rocks of the Qiangtang block.

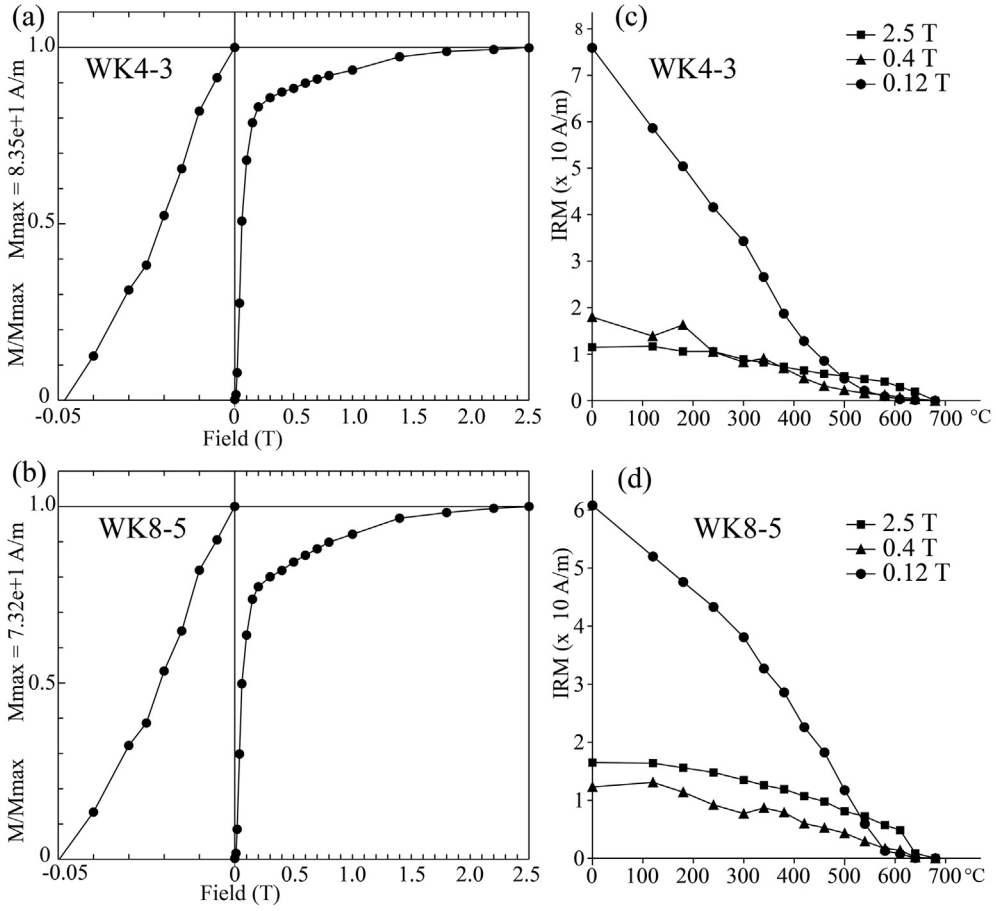


Fig. 5. Results of (a, b) the SIRM acquisition and the back-field demagnetization curves and (c, d) the thermal demagnetization of a three-component IRM of representative samples of the Early Cretaceous volcanic rocks of the Qiangtang block.

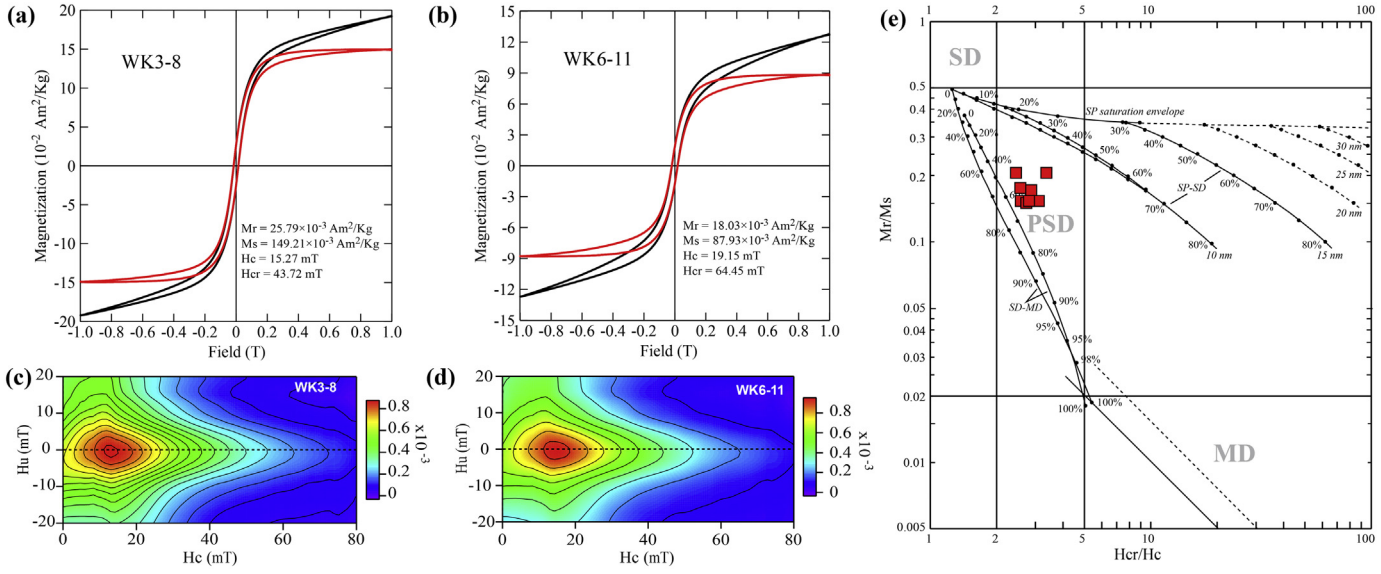


Fig. 6. (a-b) Hysteresis loops, (c-d) first-order reversal curve (FORC) diagrams, and (e) Day plot (Dunlop, 2002) of the hysteresis parameters of representative samples of the Early Cretaceous volcanic rocks of the Qiangtang block. The red lines in (a) and (b) are the hysteresis loops with the diamagnetic correction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

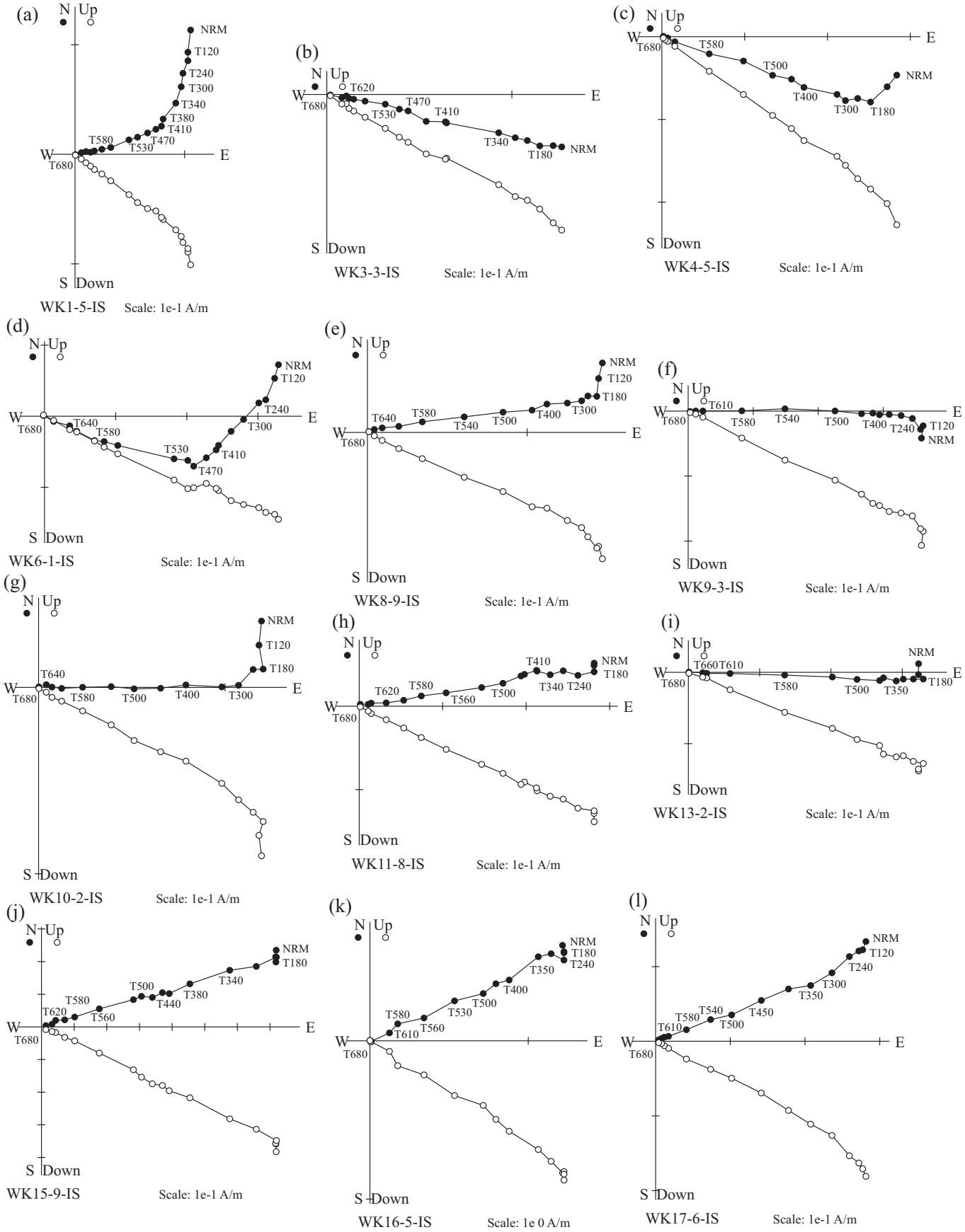


Fig. 7. Orthogonal demagnetization diagrams (in situ coordinates) for representative samples of the Early Cretaceous volcanic rocks of the Qiangtang block. Demagnetization steps in °C in all plots. NRM: natural remanent magnetization. Open (closed) dots represent vertical (horizontal) projections.

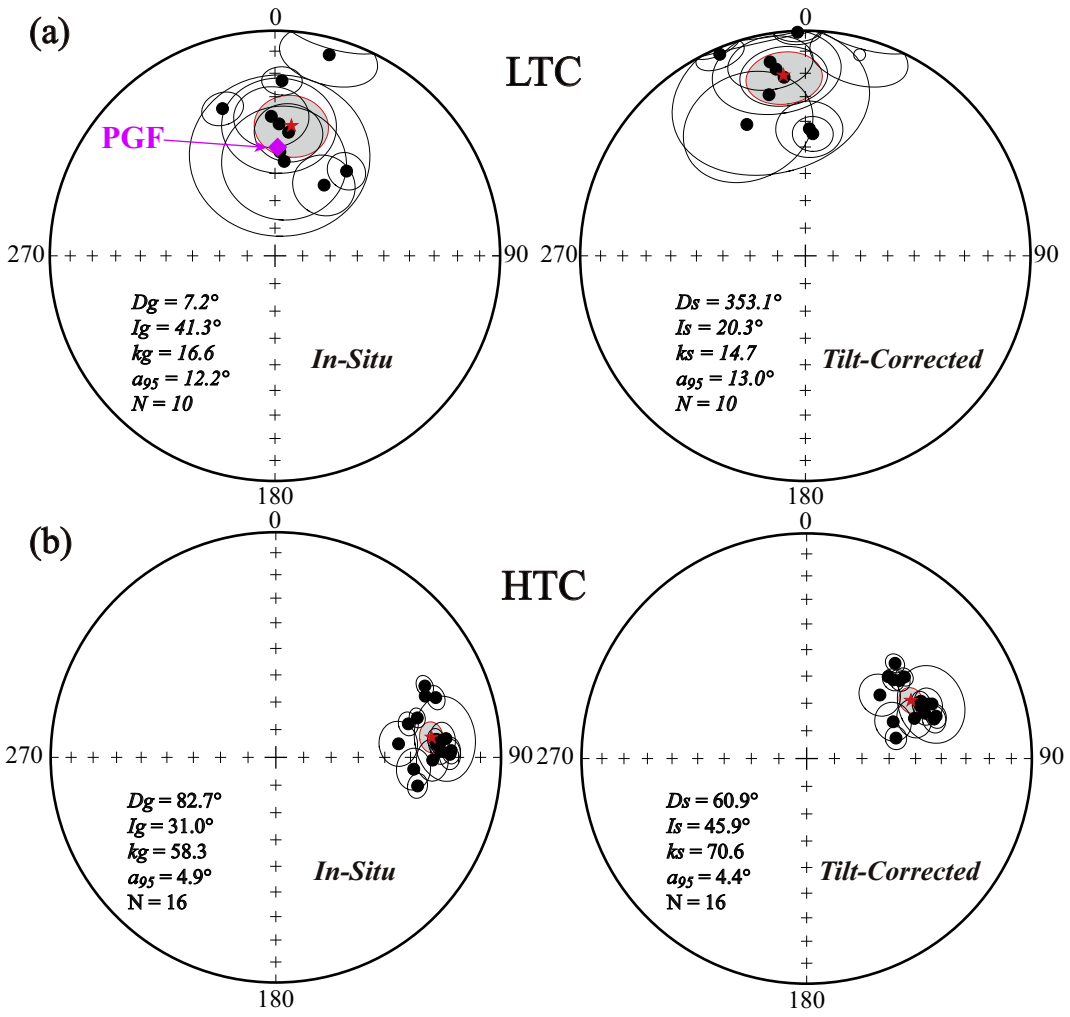


Fig. 8. Equal-area projections of the site-mean direction of the (a) LTC and (b) HTC of the Early Cretaceous volcanic rocks of the Qiangtang block. Solid/open symbols represent downward/upward inclinations.

Table 1
Site-mean paleomagnetic directions of the HTC of the Early Cretaceous volcanic rocks of the Qiangtang block.

Sites	Strike/dip (°)	n/N	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95} (°)	Plat. (°N)	Plon. (°E)	dp/dm (°)
WK1	228/30	6/9	74.4	35.1	50.2	42.8	441.5	3.2	45.7	170.4	2.4/4.0
WK2	228/30	8/10	83.7	44.4	49.1	54.5	55.7	7.5	49.6	156.4	7.5/10.6
WK3	228/30	7/8	94.9	38.5	67.0	55.6	79.4	6.8	35.8	150.0	6.9/9.7
WK4	228/30	7/8	101.4	36.0	77.1	56.5	224.8	4.0	28.5	145.7	4.2/5.8
WK5	228/30	7/8	75.8	39.2	47.8	46.8	234.5	3.9	48.9	167.2	3.2/5.0
WK6	228/30	5/8	83.7	25.0	66.5	39.1	35.6	13.0	31.0	165.8	9.3/15.5
WK7	228/30	No reliable site-mean direction									
WK8	228/30	10/10	87.8	27.6	68.9	43.2	141.1	4.1	30.3	161.5	3.2/5.1
WK9	228/30	10/10	89.0	23.7	72.8	40.4	111.3	4.6	26.2	161.9	3.4/5.6
WK10	228/30	8/11	85.4	29.0	65.4	43.3	270.5	3.4	33.2	162.9	2.6/4.2
WK11	228/30	8/9	83.6	29.3	63.4	42.6	266.1	3.4	34.6	164.4	2.6/4.2
WK12	228/30	11/11	91.0	31.0	69.6	47.6	49.5	6.6	31.2	157.4	5.6/8.6
WK13	228/30	6/8	84.0	26.4	65.9	40.4	154.4	5.4	31.9	165.1	3.9/6.5
WK14	228/30	8/8	87.8	23.1	71.9	39.3	190.1	4.0	26.6	163.1	2.9/4.8
WK15	204/33	9/9	69.5	25.2	50.0	45.1	186.3	3.8	46.6	168.1	3.0/4.8
WK16	204/33	8/9	67.7	29.1	44.9	47.4	1361.0	1.5	51.4	167.7	1.3/1.9
WK17	209/33	9/9	64.4	27.5	42.9	41.8	233.4	3.4	51.6	175.4	2.5/4.2
Mean		16(127)	82.7	31.0			58.3	4.9	37.9	162.2	k = 53.5
					60.9	45.9	70.6	4.4			A_{95} = 5.1

Notes: n/N: number of samples used to calculate mean directions/total samples demagnetized; Dg, Ig: declination and inclination in geographic coordinate; Ds, Is: declination and inclination in stratigraphic coordinate; k: the best estimate of the precision parameter; α_{95}: the radius that the mean direction lies within 95% confidence; Plat. and Plon., latitude and longitude of paleomagnetic poles; dp/dm: semi-axes of elliptical error of the pole at a probability of 95%; A_{95}: the radius that the mean pole lies within 95% confidence.

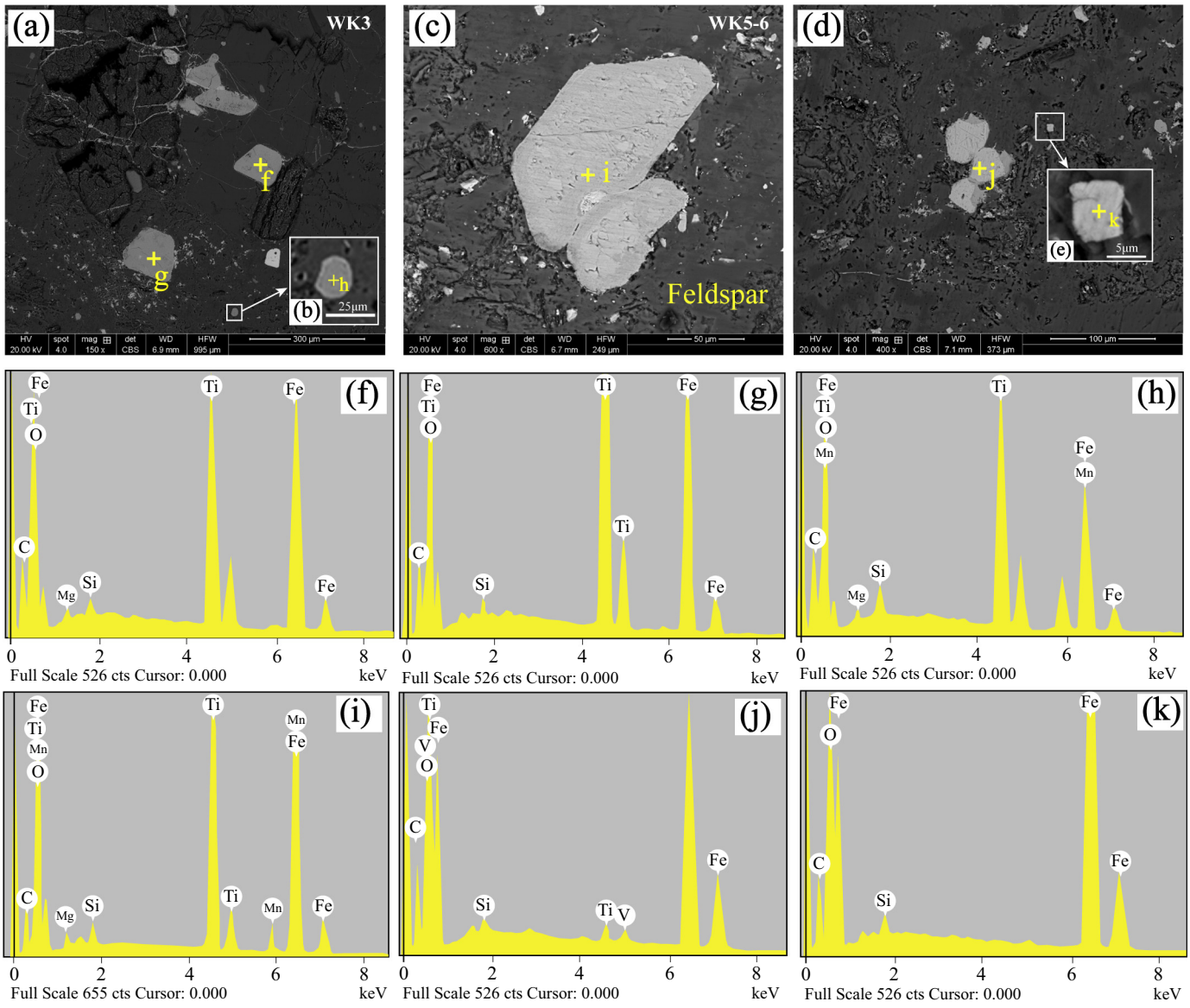


Fig. 9. (a-e) SEM backscattered electron images and (f-k) the results of the EDS analyses of representative samples of the Early Cretaceous volcanic rocks of the Qiangtang block.

primary remanence of the Early Cretaceous (~120–115 Ma) and is suitable for tectonic analysis for the Qiangtang block.

5.2. Cretaceous paleolatitude of the Qiangtang block

The available Cretaceous studies of the Qiangtang block are listed in Table 2. In this study, Early Cretaceous (~120–115 Ma) paleomagnetic results for the western portion of the Qiangtang block have been obtained. The paleomagnetic direction is Ds/Is = 60.9°/45.9° with α_{95} = 4.4°, and the corresponding paleopole lies at 37.9°N, 162.2°E with A_{95} = 5.1°, indicating a paleolatitude of 27.6 ± 5.0°N for the study area at ~120–115 Ma (reference point: 32.9°N, 83.5°E).

The paleomagnetic data were obtained from the ~110 Ma to ~100 Ma volcanic rocks in the western part of the Qiangtang block (Chen et al., 2017). However, the directions of sites GZ35–36, GZ37, GZ38, and GZ39–41 are significantly different from those of the other 10 sites. Meng et al. (2018) reported paleomagnetic results for the Late Cretaceous Abushan Formation in the Qiangtang block. They determined that the HTC directions had been affected by the shallowing process and used the ITC data to calculate the paleolatitude of the Qiangtang block. However, the nonparametric fold test (Tauxe and Watson, 1994) indicates that the ITC directions may represent synfolding magnetization, while the HTC directions represent prefolding magnetization. Therefore, there are significant limitations in discussing the Cretaceous paleolatitude evolution of the Qiangtang block based on existing paleomagnetic studies. Higher quality paleomagnetic data with accurate geochronologic age are needed to resolve this issue.

5.3. Insights into the timing of the Qiangtang-Lhasa collision

Reconstructing the paleolatitudinal positions of the Qiangtang and the Lhasa blocks during the Cretaceous is very important for understanding the history of the Qiangtang-Lhasa collision. Previous published Early Cretaceous paleomagnetic results for the Lhasa block are listed in Table 2, and the corresponding sampling locations are shown in Fig. 1. Three Early Cretaceous paleomagnetic poles from volcanic rocks with accurate geochronological constraints were obtained from the western part of the Lhasa block (Chen et al., 2012; Ma et al., 2014; Bian et al., 2017). These paleomagnetic poles are aligned on a well-defined small circle centered on reference point 32.5°N, 84.0°E (Gaize

Table 2
Summary of the Early Cretaceous paleomagnetic results from the Qiangtang and Lhasa blocks.

Slat. (°N)	Slon. (°E)	Age (Ma)	Lithology	N/n	Plat. (°N)	Plon. (°E)	A_{95}(dp/dm) (°)	Paleolatitude (°N)	References
Qiangtang block-Western part									
32.5	84.3	~110–100	Lava	10/57	75.8	321.9	3.8/6.3	24.2 ± 4.9	Chen et al., 2017
32.9	83.5	~120–115	Lava	16/127	37.9	162.2	5.1	27.6 ± 5.0	This study
Qiangtang block-Eastern part									
29.7	98.4	K_1	Red beds	12/56	40.6	170.5	13.0	31.7 ± 11.7	Huang et al., 1992
29.7	98.6	K_1	Red beds	5/30	48.5	175.9	9.5	29.8 ± 8.9	Otofuji et al., 1990
Lhasa block-Western part									
32.2	80.4	~116–113	Lava + Limestone	19/205	69.1	319.8	4.8	20.1 ± 5.4	Bian et al., 2017
31.1	81.4	~121–117	Lava	12/116	70.5	292.9	7.4	14.1 ± 9.3	Yang et al., 2015
32.3	82.6	~132–120	Lava	51/444	61.4	192.9	2.1	19.2 ± 2.4	Ma et al., 2014
31.3	85.1	~130–110	Lava	18/162	58.2	341.9	4.6	19.8 ± 5.2	Chen et al., 2012
Lhasa block-Middle part									
30.5	90.1	~114	Lava	15/88	66.4	220.3	6.9/9.3	14.0 ± 8.6	Sun et al., 2008
31.3	91.9	~120	Lava	19/139	66.9	281.2	6.1	13.1 ± 2.7	Li et al., 2017

Notes: Slat., geographic latitude of the studied sites; Slon., geographic longitude of the studied sites; N, number of sampling sites used for Fisher mean; n, number of samples used for Fisher mean; Plat., the latitude of the paleomagnetic poles; Plon., the longitude of the paleomagnetic poles; A_{95}, the radius of cone at 95% confidence level of paleomagnetic pole; Paleolatitude, the paleolatitudes calculated from each sampling sites.

area, western part of the BNSZ). The small circle fitting results of the Early Cretaceous (~130 Ma to ~110 Ma) paleomagnetic results indicate a paleolatitude of 20.1 ± 2.4°N for the northern margin of the western part of the Lhasa block. When compared with the results of the Early Cretaceous (~120–115 Ma) paleomagnetic data from the western part of the Qiangtang block (this study), a discrepancy in the paleolatitudes (7.5° ± 5.5°) of the western parts of the Qiangtang and Lhasa blocks can be inferred (Fig. 10a).

The observed paleolatitudinal difference of the western Qiangtang and Lhasa blocks during the Early Cretaceous (~120–115 Ma) can be attributed to crustal shortening or to a spatial gap between the two blocks. However, we favor the spatial gap between the western parts of the Qiangtang and Lhasa blocks based on the following lines of evidence. First, geologic mapping and structural restoration suggest that approximately 50% to 60% of N-S crustal shortening occurred during the Cretaceous in the western and middle parts of the Lhasa block (Murphy et al., 1997; Kapp et al., 2005, 2007; Volkmer et al., 2007). According to the estimates of crustal shortening, only ~100 km of N-S convergence is calculated to have occurred between the study area and the sampling sites in the western part of the Lhasa block (Chen et al., 2012; Ma et al., 2014; Bian et al., 2017) at the longitude of the sampling sites (~84°E) since the Cretaceous. This is much less than the observed paleolatitudinal difference. Second, GPS and geological studies of the surface rupture have shown that the strike slip fault system has experienced a limited amount of slip within the western-middle parts of the Qiangtang block (Han et al., 2019). It seems that the western part of the Qiangtang block did not undergo obvious lateral extrusion. Finally, although northward underthrusting caused the Lhasa block to subduct

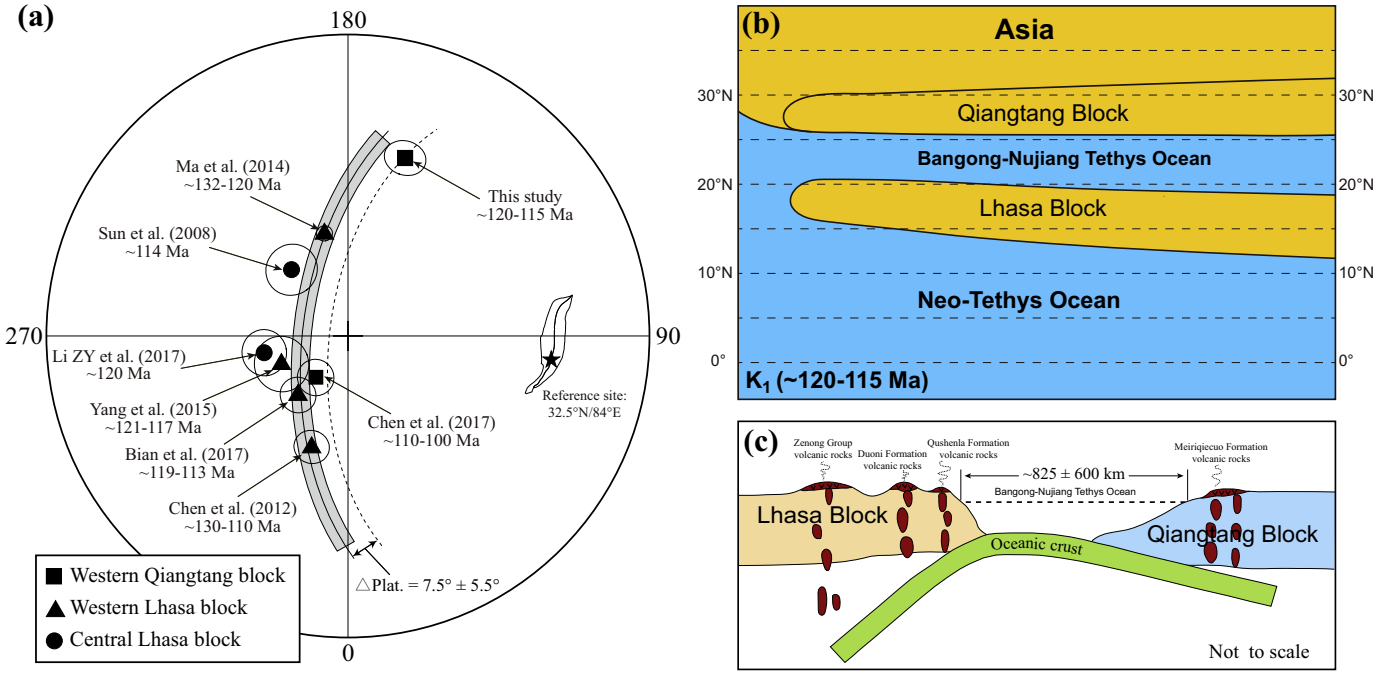


Fig. 10. (a) Equal-area projection showing the available Early Cretaceous paleomagnetic poles of the Qiangtang and Lhasa blocks. (b) Early Cretaceous (~120–115 Ma) paleogeographic reconstruction of the western parts of the Qiangtang and Lhasa blocks. (c) Cross-sections of the western parts of the Qiangtang and Lhasa blocks during the Early Cretaceous (~120–115 Ma) (modified from Li et al., 2015).

beneath the Qiantang block (Kapp et al., 2005; Volkmer et al., 2007), geophysical data show a limited distance (~50–100 km) for this subduction (Li et al., 2006; Gao et al., 2013). Therefore, we conclude that the paleolatitudinal difference implies that there was a spatial gap between the western parts of the Qiangtang and Lhasa blocks at 120–115 Ma. This indicates that the extent of the western segment of the BNTO was ~825 ± 600 km (7.5° ± 5.5° in latitude) at 120–115 Ma (Fig. 10) and that the collision between the western parts of the Qiangtang and Lhasa blocks occurred later than 115 Ma.

The collision time is also supported by the existing geological evidence. (1) The youngest marine sedimentary strata exposed in the western part of the Qiangtang block is Early Cretaceous. For example, U−Pb ages of the basalts interbedded in the marine sedimentary strata (Xueshan Formation) of the western part of the Qiangtang block is 118 ± 2.1 Ma (Bi et al., 2018), that is, Early Cretaceous. (2) The Early Cretaceous ophiolite distributed along the western BNSZ. The ages constrained by isotope chronology and/or radiolarian dating of the ophiolites in Julu, Geiji, Zhongcang, and Dongcuo are 103.8 ± 3.9 Ma (Liu et al., 2014), 126.6 ± 10.2 Ma (Zhang et al., 2014), 114.3 ± 1.4 Ma (Xu et al., 2014), and 131–121 Ma (Baxter et al., 2009), respectively. (3) The presence of Early Cretaceous magmatism is related to the slab subduction of the BNTO. The Meiriqiecuo Formation volcanic rocks (~120–110 Ma) (Ma et al., 2015; Wei et al., 2017) and the granitoid rocks (~123.7–107.3 Ma) in the Chem Co area (Han et al., 2020) of the western part of the Qiangtang block were formed as a result of the northward subduction of the BNTO. (4) The Early Cretaceous ocean island rock assemblages are widely distributed within the BNSZ. The geochronologic data indicate that the Zhonggang ocean island, located in the western part of the BNTO, formed during the Early Jurassic to Early Cretaceous (185–116 Ma) (Fan et al., 2018). These findings indicate that the BNTO did not close until the middle of the Late Cretaceous and the width of the ocean was still considerable during this time.

The timing of the Qiangtang-Lhasa collision remains contentious, and the proposed age of the Qiangtang-Lhasa collision ranges from the Middle Jurassic to the Late Cretaceous. The paleomagnetic results obtained in this study suggest that the collisions of the western parts of the Qiangtang and Lhasa blocks occurred later than 115 Ma. This study provides quantitative data for understanding the collisional process between the Qiangtang and Lhasa blocks and the evolution of the central Tibetan Plateau.

6. Conclusions

This study provides new reliable paleomagnetic results for the Early Cretaceous volcanic rocks in the western part of the Qiangtang block. These results suggest that the western part of the Qiangtang block was located at 27.6 ± 5.0°N at 120–115 Ma (reference point: 32.9°N, 83.5°E). By comparing the new paleomagnetic results and the available Cretaceous paleomagnetic data from the Qiangtang and Lhasa blocks, we drew the following conclusions: (1) The extent of the western segment of the BNTO was ~825 ± 600 km (7.5° ± 5.5° in latitude) at 120–115 Ma. (2) The final closure of the BNTO and the collision of the western parts of the Qiangtang and Lhasa blocks occurred later than 115 Ma.

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Credit Author Statement.

Yong Cao: Conceptualization, Methodology, Data curation, Investigation, Funding acquisition, Writing - Original Draft,

Zhiming Sun: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Supervision, Funding acquisition, Writing - Review & Editing.

Haibing Li: Conceptualization, Formal analysis, Supervision.

Xiaozhou Ye: Methodology, Data curation, Investigation,

Jiawei Pan: Investigation, Funding acquisition.

Dongliang Liu: Methodology, Resources.

Lei Zhang: Methodology, Resources.

Bailing Wu: Methodology, Data curation.

Xinwen Cao: Methodology, Data curation.

Chenguang Liu: Methodology, Data curation.

Zhenyu Yang: Conceptualization.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tecto.2020.228500.

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